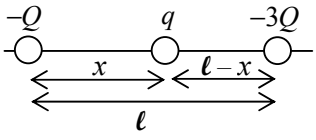
$$Q_1 = \frac{Q \pm \sqrt{Q^2 - 4\dfrac{Fd^2}{k}}}{2}$$

$$= \tfrac{1}{2}\left[(90.0\times10^{-6}\text{ C}) \pm \sqrt{(90.0\times10^{-6}\text{ C})^2 - 4\frac{(-12.0\text{ N})(0.280\text{ m})^2}{(8.988\times10^{9}\text{ N}\cdot\text{m}^2/\text{C}^2)}} \right]$$

$$= \boxed{91.1\times10^{-6}\text{ C}, -1.1\times10^{-6}\text{ C}}$$

18. The negative charges will repel each other, so the third charge must put an opposite force on each of the original charges. Consideration of the various possible configurations leads to the conclusion that the third charge must be positive and must be between the other two charges. See the diagram for the definition of variables. For each negative charge, equate the magnitudes of the two forces on the charge. Also note that $0 < x < \ell$.



$$\text{Left: } k\frac{Qq}{x^2} = k\frac{3Q^2}{\ell^2} \qquad \text{Right: } k\frac{3Qq}{(\ell - x)^2} = k\frac{3Q^2}{\ell^2} \quad \rightarrow$$

$$k\frac{Qq}{x^2} = k\frac{3Qq}{(\ell - x)^2} \quad \rightarrow \quad x = \frac{\ell}{\sqrt{3}+1} = 0.366\ell$$

$$k\frac{Qq}{x^2} = k\frac{3Q^2}{\ell^2} \quad \rightarrow \quad q = 3Q\frac{x^2}{\ell^2} = Q\frac{3}{\left(\sqrt{3}+1\right)^2} = 0.402Q$$

Thus the charge should be of magnitude $\boxed{0.40\,Q}$, and a distance $\boxed{0.37\ell \text{ from} -Q \text{ toward} -3Q}$.

19. Use Eq. 16–3 to calculate the force.

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q} \quad \rightarrow \quad \vec{\mathbf{F}} = q\vec{\mathbf{E}} = (-1.602\times10^{-19}\text{ C})(2460\text{ N/C east}) = \boxed{3.94\times10^{-16}\text{ N west}}$$

20. Use Eq. 16–3 to calculate the electric field.

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q} = \frac{1.86\times10^{-14}\text{ N south}}{1.602\times10^{-19}\text{ C}} = \boxed{1.16\times10^{5}\text{ N/C south}}$$

21. Use Eq. 16–4a to calculate the electric field due to a point charge.

$$E = k\frac{Q}{r^2} = (8.988\times10^{9}\text{ N}\cdot\text{m}^2/\text{C}^2)\frac{33.0\times10^{-6}\text{ C}}{(0.217\text{ m})^2} = \boxed{6.30\times10^{6}\text{ N/C upward}}$$

Note that the electric field points away from the positive charge.

22. Use Eq. 16–3 to calculate the electric field.

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q} = \frac{6.4\text{ N down}}{-7.3\times10^{-6}\text{ C}} = \boxed{8.8\times10^{5}\text{ N/C up}}$$

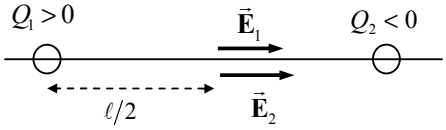
© Copyright 2014 Pearson Education, Inc. All rights reserved. This material is protected under all copyright laws as they currently exist. No portion of this material may be reproduced, in any form or by any means, without permission in writing from the publisher.

23. Assuming the electric force is the only force on the electron, then Newton's second law may be used with Eq. 16–5 to find the acceleration.

$$\vec{\mathbf{F}}_{\text{net}} = m\vec{\mathbf{a}} = q\vec{\mathbf{E}} \quad \rightarrow \quad a = \frac{|q|}{m}E = \frac{(1.602\times10^{-19}\text{ C})}{(9.11\times10^{-31}\text{ kg})}756\text{ N/C} = \boxed{1.33\times10^{14}\text{ m/s}^2}$$

Since the charge is negative, the direction of the acceleration is opposite to the field.

24. The electric field due to the positive charge (Q_1) will point away from it, and the electric field due to the negative charge (Q_2) will point toward it. Thus both fields point in the same direction, toward the negative charge, and the magnitudes can be added.

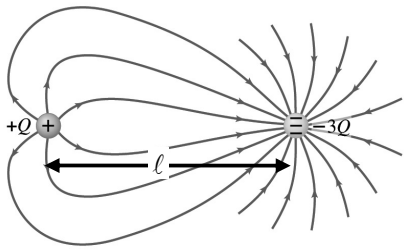


$$E = |E_1| + |E_2| = k\frac{|Q_1|}{r_1^2} + k\frac{|Q_2|}{r_2^2} = k\frac{|Q_1|}{(\ell/2)^2} + k\frac{|Q_2|}{(\ell/2)^2} = \frac{4k}{\ell^2}(|Q_1| + |Q_2|)$$

$$= \frac{4(8.988\times10^9\text{ N}\cdot\text{m}^2/\text{C}^2)}{(0.060\text{ m})^2}(8.0\times10^{-6}\text{ C} + 5.8\times10^{-6}\text{ C}) = 1.3782\times10^8\text{ N/C} \approx \boxed{1.4\times10^8\text{ N/C}}$$

The direction is toward the negative charge.

25.



26. Assuming the electric force is the only force on the electron, Newton's second law may be used to find the electric field strength.

$$F_{\text{net}} = ma = qE \quad \rightarrow$$

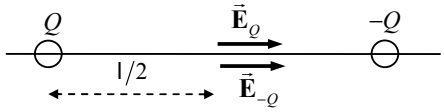
$$E = \frac{ma}{q} = \frac{(1.673\times10^{-27}\text{ kg})(2.4\times10^6)(9.80\text{ m/s}^2)}{(1.602\times10^{-19}\text{ C})} = 0.2456\text{ N/C} \approx \boxed{0.25\text{ N/C}}$$

27. Since the electron accelerates from rest toward the north, the net force on it must be to the north. Assuming the electric force is the only force on the electron, Newton's second law and Eq. 16–5 may be used to find the electric field.

$$\vec{\mathbf{F}}_{\text{net}} = m\vec{\mathbf{a}} = q\vec{\mathbf{E}} \quad \rightarrow \quad \vec{\mathbf{E}} = \frac{m}{q}\vec{\mathbf{a}} = \frac{(9.11\times10^{-31}\text{ kg})}{(-1.602\times10^{-19}\text{ C})}(105\text{ m/s}^2\text{ north}) = \boxed{5.97\times10^{-10}\text{ N/C south}}$$

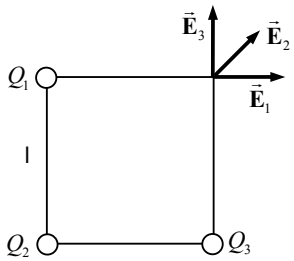
© Copyright 2014 Pearson Education, Inc. All rights reserved. This material is protected under all copyright laws as they currently exist. No portion of this material may be reproduced, in any form or by any means, without permission in writing from the publisher.

28. The field due to the negative charge will point toward the negative charge, and the field due to the positive charge will point toward the negative charge. Thus the magnitudes of the two fields can be added together to find the charges.



$$E_{\text{net}} = 2E_Q = 2k\frac{Q}{(\ell/2)^2} = \frac{8kQ}{\ell^2} \quad \rightarrow \quad Q = \frac{E\ell^2}{8k} = \frac{(386\text{ N/C})(0.160\text{ m})^2}{8(8.988\times10^9\text{ N}\cdot\text{m}^2/\text{C}^2)} = \boxed{1.37\times10^{-10}\text{ C}}$$

29. The field at the upper right-hand corner of the square is the vector sum of the fields due to the other three charges. Let ℓ represent the 1.22-m length of a side of the square, and let Q represent the charge at each of the three occupied corners.



$$E_1 = k\frac{Q}{\ell^2} \quad \rightarrow \quad E_{1x} = k\frac{Q}{\ell^2},\ E_{1y} = 0$$

$$E_2 = k\frac{Q}{2\ell^2} \quad \rightarrow \quad E_{2x} = k\frac{Q}{2\ell^2}\cos 45° = k\frac{\sqrt{2}Q}{4\ell^2},\ E_{2y} = k\frac{\sqrt{2}Q}{4\ell^2}$$

$$E_3 = k\frac{Q}{\ell^2} \quad \rightarrow \quad E_{3x} = 0,\ E_{1y} = k\frac{Q}{\ell^2}$$

Add the x and y components together to find the total electric field, noting that $E_x = E_y$.

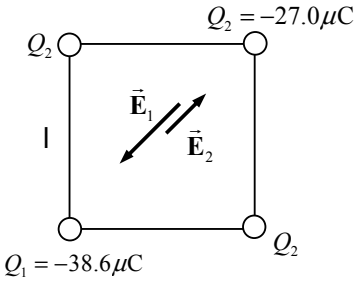
$$E_x = E_{1x} + E_{2x} + E_{3x} = k\frac{Q}{\ell^2} + k\frac{\sqrt{2}Q}{4\ell^2} + 0 = k\frac{Q}{\ell^2}\left(1+\frac{\sqrt{2}}{4}\right) = E_y$$

$$E = \sqrt{E_x^2 + E_y^2} = k\frac{Q}{\ell^2}\left(1+\frac{\sqrt{2}}{4}\right)\sqrt{2} = k\frac{Q}{\ell^2}\left(\sqrt{2}+\frac{1}{2}\right)$$

$$= (8.988\times10^9\text{ N}\cdot\text{m}^2/\text{C}^2)\frac{(3.25\times10^{-6}\text{ C})}{(1.22\text{ m})^2}\left(\sqrt{2}+\frac{1}{2}\right) = \boxed{3.76\times10^4\text{ N/C}}$$

$$\theta = \tan^{-1}\frac{E_y}{E_x} = \boxed{45.0°} \text{ from the } x \text{ direction}$$

30. The fields at the center due to the two $-27.0\ \mu\text{C}$ negative charges on opposite corners (lower right and upper left in the diagram) will cancel each other, so only the other two charges need to be considered. The field due to each of the other charges will point directly toward its source charge. Accordingly, the two fields are in opposite directions and can be combined algebraically. The distance from each charge to the center is $\ell/\sqrt{2}$.



© Copyright 2014 Pearson Education, Inc. All rights reserved. This material is protected under all copyright laws as they currently exist. No portion of this material may be reproduced, in any form or by any means, without permission in writing from the publisher.

$$E = E_1 - E_2 = k\frac{|Q_1|}{\ell^2/2} - k\frac{|Q_2|}{\ell^2/2} = k\frac{|Q_1|-|Q_2|}{\ell^2/2} \quad = (8.988\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)\frac{(38.6-27.0)\times10^{-6}\ \text{C}}{(0.425\ \text{m})^2/2}$$

$$= \boxed{1.15\times10^6\ \text{N/C, toward the} - 38.6\ \mu\text{C charge}}$$

31. Choose the rightward direction to be positive. Then the field due to $+Q$ will be positive, and the field due to $-Q$ will be negative.

$$E = k\frac{Q}{(x+a)^2} - k\frac{Q}{(x-a)^2} = kQ\left(\frac{1}{(x+a)^2} - \frac{1}{(x-a)^2}\right) = \boxed{\frac{-4kQxa}{(x^2-a^2)^2}}$$

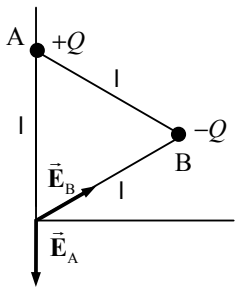
The negative sign means the field points to the left.

32. For the net field to be zero at point P, the magnitudes of the fields created by Q_1 and Q_2 must be equal. Also, the distance x will be taken as positive to the left of Q_1. That is the only region where the total field due to the two charges can be zero. Let the variable ℓ represent the 12-cm distance.

$$\left|\vec{\mathbf{E}}_1\right| = \left|\vec{\mathbf{E}}_2\right| \quad \rightarrow \quad k\frac{|Q_1|}{x^2} = k\frac{Q_2}{(x+\ell)^2} \quad \rightarrow$$

$$x = \ell\frac{\sqrt{|Q_1|}}{\left(\sqrt{Q_2} - \sqrt{|Q_1|}\right)} = (12\ \text{cm})\frac{\sqrt{32\mu\text{C}}}{\left(\sqrt{45\mu\text{C}} - \sqrt{32\mu\text{C}}\right)} = 64.57\ \text{cm} \approx \boxed{65\ \text{cm}}$$

33. The field due to the charge at A will point straight downward, and the field due to the charge at B will point along the line from the origin to point B, 30° above the x axis. We first solve the problem symbolically and then substitute in the values.



$$E_A = k\frac{Q}{\ell^2} \quad \rightarrow \quad E_{Ax} = 0,\ E_{Ay} = -k\frac{Q}{\ell^2}$$

$$E_B = k\frac{Q}{\ell^2} \quad \rightarrow \quad E_{Bx} = k\frac{Q}{\ell^2}\cos 30° = k\frac{\sqrt{3}Q}{2\ell^2},$$

$$E_{By} = k\frac{Q}{\ell^2}\sin 30° = k\frac{Q}{2\ell^2}$$

$$E_x = E_{Ax} + E_{Bx} = k\frac{\sqrt{3}Q}{2\ell^2} \qquad E_y = E_{Ay} + E_{By} = -k\frac{Q}{2\ell^2}$$

$$E = \sqrt{E_x^2 + E_y^2} = \sqrt{\frac{3k^2Q^2}{4\ell^4} + \frac{k^2Q^2}{4\ell^4}} = \sqrt{\frac{4k^2Q^2}{4\ell^4}} = \frac{kQ}{\ell^2}$$

$$= \frac{(8.988\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(26\times10^{-6}\ \text{C})}{(0.080\ \text{m})^2} = 3.651\times10^7\ \text{N/C} \approx \boxed{3.7\times10^7\ \text{N/C}}$$

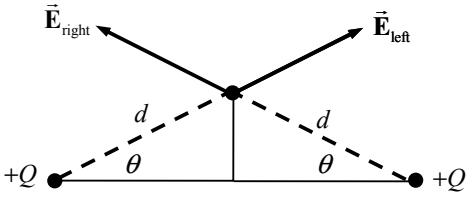
$$\theta = \tan^{-1}\frac{E_y}{E_x} = \tan^{-1}\frac{-k\dfrac{Q}{2\ell^2}}{k\dfrac{\sqrt{3}Q}{2\ell^2}} = \tan^{-1}\frac{-1}{\sqrt{3}} = \boxed{330°}$$

© Copyright 2014 Pearson Education, Inc. All rights reserved. This material is protected under all copyright laws as they currently exist. No portion of this material may be reproduced, in any form or by any means, without permission in writing from the publisher.

34. The charges must be of the same sign so that the electric fields created by the two charges oppose each other and add to zero. The magnitudes of the two electric fields must be equal.

$$E_1 = E_2 \quad \rightarrow \quad k\frac{Q_1}{(\ell/3)^2} = k\frac{Q_2}{(2\ell/3)^2} \quad \rightarrow \quad 9Q_1 = \frac{9Q_2}{4} \quad \rightarrow \quad \frac{Q_1}{Q_2} = \boxed{\frac{1}{4}}$$

35. In each case, find the vector sum of the field caused by the charge on the left ($\vec{\mathbf{E}}_{\text{left}}$) and the field caused by the charge on the right ($\vec{\mathbf{E}}_{\text{right}}$).



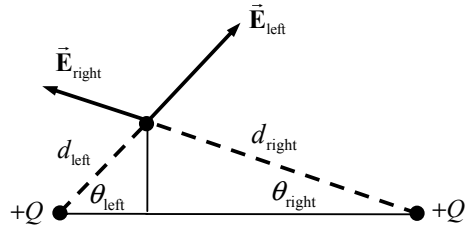
Point A: From the symmetry of the geometry, in calculating the electric field at point A, only the vertical components of the fields need to be considered. The horizontal components will cancel each other.

$$\theta = \tan^{-1}\frac{5.0}{10.0} = 26.6°$$

$$d = \sqrt{(0.050\text{ m})^2 + (0.100\text{ m})^2} = 0.1118\text{ m}$$

$$E_A = 2\frac{kQ}{d^2}\sin\theta = 2(8.988\times10^9\text{ N}\cdot\text{m}^2/\text{C}^2)\frac{4.7\times10^{-6}\text{C}}{(0.1118\text{ m})^2}\sin 26.6° = \boxed{3.0\times10^6\text{ N/C}} \quad \theta_A = \boxed{90°}$$

Point B: Now the point is not symmetrically placed. The horizontal and vertical components of each individual field need to be calculated to find the resultant electric field.



$$\theta_{\text{left}} = \tan^{-1}\frac{5.0}{5.0} = 45° \qquad \theta_{\text{left}} = \tan^{-1}\frac{5.0}{15.0} = 18.4°$$

$$d_{\text{left}} = \sqrt{(0.050\text{ m})^2 + (0.050\text{ m})^2} = 0.0707\text{ m}$$

$$d_{\text{right}} = \sqrt{(0.050\text{ m})^2 + (0.150\text{ m})^2} = 0.1581\text{ m}$$

$$E_x = (\vec{\mathbf{E}}_{\text{left}})_x + (\vec{\mathbf{E}}_{\text{right}})_x = k\frac{Q}{d_{\text{left}}^2}\cos\theta_{\text{left}} - k\frac{Q}{d_{\text{right}}^2}\cos\theta_{\text{right}}$$

$$= (8.988\times10^9\text{ N}\cdot\text{m}^2/\text{C}^2)(4.7\times10^{-6}\text{ C})\left[\frac{\cos 45°}{(0.0707\text{ m})^2} - \frac{\cos 18.4°}{(0.1581\text{ m})^2}\right] = 4.372\times10^6\text{ N/C}$$

$$E_y = (\vec{\mathbf{E}}_{\text{left}})_y + (\vec{\mathbf{E}}_{\text{right}})_y = k\frac{Q}{d_{\text{left}}^2}\sin\theta_{\text{left}} + k\frac{Q}{d_{\text{right}}^2}\sin\theta_{\text{right}}$$

$$= (8.988\times10^9\text{ N}\cdot\text{m}^2/\text{C}^2)(4.7\times10^{-6}\text{ C})\left[\frac{\sin 45°}{(0.0707\text{ m})^2} + \frac{\sin 18.4°}{(0.1581\text{ m})^2}\right] = 6.509\times10^6\text{ N/C}$$

$$E_B = \sqrt{E_x^2 + E_y^2} = 7.841\times10^6\text{ N/C} \approx \boxed{7.8\times10^6\text{ N/C}} \qquad \theta_B = \tan^{-1}\frac{E_y}{E_x} = 56.11° \approx \boxed{56°}$$

© Copyright 2014 Pearson Education, Inc. All rights reserved. This material is protected under all copyright laws as they currently exist. No portion of this material may be reproduced, in any form or by any means, without permission in writing from the publisher.

The results are consistent with Fig. 16–32b. In the figure, the field at point A points straight up, matching the calculations. The field at point B should be to the right and vertical, matching the calculations. Finally, the field lines are closer together at point B than at point A, indicating that the field is stronger there, again matching the calculations.

36. We assume that gravity can be ignored, which is proven in part (*b*).

(*a*) The electron will accelerate to the right. The magnitude of the acceleration can be found from setting the net force equal to the electric force on the electron. The acceleration is constant, so constant-acceleration relationships can be used.

$$F_{\text{net}} = ma = |q|E \quad \rightarrow \quad a = \frac{|q|E}{m}$$

$$v^2 = v_0^2 + 2a\Delta x \quad \rightarrow \quad v = \sqrt{2a\Delta x} = \sqrt{2\frac{|q|E}{m}\Delta x}$$

$$= \sqrt{2\frac{(1.602\times10^{-19}\text{ C})(1.45\times10^{4}\text{ N/C})}{(9.11\times10^{-31}\text{ kg})}(1.60\times10^{-2}\text{ m})} = \boxed{9.03\times10^{6}\text{ m/s}}$$

(*b*) The value of the acceleration caused by the electric field is compared to *g*.

$$a = \frac{|q|E}{m} = \frac{(1.602\times10^{-19}\text{ C})(1.45\times10^{4}\text{ N/C})}{(9.11\times10^{-31}\text{ kg})} = 2.55\times10^{15}\text{ m/s}^2$$

$$a/g = (2.55\times10^{15}\text{ m/s}^2)/(9.80\text{ m/s}^2) = 2.60\times10^{14}$$

The acceleration due to gravity can be ignored compared to the acceleration caused by the electric field.

37. (*a*) The net force between the thymine and adenine is due to the following forces.

O–H attraction: $F_{\text{OH}} = k\dfrac{(0.4e)(0.2e)}{(1.80\times10^{-10}\text{ m})^2} = \dfrac{0.08ke^2}{(1.80\times10^{-10}\text{ m})^2}$

O–N repulsion: $F_{\text{ON}} = k\dfrac{(0.4e)(0.2e)}{(2.80\times10^{-10}\text{ m})^2} = \dfrac{0.08ke^2}{(2.80\times10^{-10}\text{ m})^2}$

N–N repulsion: $F_{\text{NN}} = k\dfrac{(0.2e)(0.2e)}{(3.00\times10^{-10}\text{ m})^2} = \dfrac{0.04ke^2}{(3.00\times10^{-10}\text{ m})^2}$

H–N attraction: $F_{\text{HN}} = k\dfrac{(0.2e)(0.2e)}{(2.00\times10^{-10}\text{ m})^2} = \dfrac{0.04ke^2}{(2.00\times10^{-10}\text{ m})^2}$

$$F_{\text{A–T}} = F_{\text{OH}} - F_{\text{ON}} - F_{\text{NN}} + F_{\text{HN}} = \left(\frac{0.08}{1.80^2} - \frac{0.08}{2.80^2} - \frac{0.04}{3.00^2} + \frac{0.04}{2.00^2}\right)\left(\frac{1}{1.0\times10^{-10}\text{ m}}\right)^2\frac{ke^2}{d^2}$$

$$= (0.02004)\frac{(8.988\times10^{9}\text{ N}\cdot\text{m}^2/\text{C}^2)(1.602\times10^{-19}\text{ C})^2}{(1.0\times10^{-10}\text{ m})^2} = 4.623\times10^{-10}\text{ N} \approx \boxed{5\times10^{-10}\text{ N}}$$

© Copyright 2014 Pearson Education, Inc. All rights reserved. This material is protected under all copyright laws as they currently exist. No portion of this material may be reproduced, in any form or by any means, without permission in writing from the publisher.